Causal Hypergraphs and Exact Schedule Classification in Rational Pushdown Normalization

Alp Eren Bütün
Independent Researcher
github.com/alperenbutun

August 2026

Abstract

We study optional eager-normalization schedules in the canonical literal-demand realization of rational $p : q$ pushdown normalization. For arbitrary coprime positive integers p, q, we derive a uniform centered compiler and an exact finite universe of baseline-portable eager sites. The site universe has at most eight vertices, with explicit parity-dependent threshold formulas for the possible extra sites. We then determine all minimal fatal supports and classify the admissible schedule family for every coprime ratio. For $\min(p, q) \geq 3$, every subset of the canonical core is admissible and each present extra site is singleton-fatal. For $\min(p, q) = 2$, the only non-singleton obstructions are the two fork supports inherited from the low-width model, while the extra sites remain singleton-fatal. The exceptional $\min(p, q) = 1$ line is classified separately, including exact admissible-schedule counts. In every ratio, the canonical weighted-balance schedule family is the independence complex of an explicit finite hypergraph and is therefore downward closed.

We then separate this family-specific theorem from two broader structural questions. First, we give a two-site exact-lift counterexample showing that arbitrary admissible schedule families need not be downward closed: enabling a second site can repair a failure created by the first. Second, we resolve the persistence of the classical $2 : 1$ causal edge $A \to B$ by distinguishing lift data from schedule-presentation data. The edge is removable when the optional-site menu may vary within the same literal one-top policy, but persists when the designated sites and a finite causal certificate are preserved. Thus causal persistence is genuinely category-dependent.

Keywords. deterministic pushdown automata; rational balance languages; schedule classification; causal hypergraphs; shortest completion; exact lifts.

1 Introduction

The real-time pushdown normalization of the rational balance language

$$L_{p,q} = \{w \in \{0, 1\}^* : qN_1(w) = pN_0(w)\}, \qquad \gcd(p, q) = 1,$$

contains a rigid mandatory normalization layer and, in selected local configurations, optional eager returns to the canonical center representation. Once those optional returns are regarded as a schedule menu, a finite combinatorial problem appears: which subsets of optional rules preserve the language, which subsets fail, and what is the causal structure of the failure?

The original $2 : 1$ construction exhibits a small but nontrivial schedule geometry. Subsequent $3 : 2$ and odd-p:2 examples show that the same phenomenon persists under rational generalization, but those examples do not by themselves determine the arbitrary-coprime pattern. The first goal of this paper is therefore exact: identify the canonical optional-site universe and classify every admissible schedule for every coprime $p : q$.

The second goal is conceptual. The exact rational family turns out to have a downward-closed admissible set, hence an independence-complex description by minimal fatal supports. We show that this is not a universal theorem of exact pushdown lifts. Finally, we address a typing issue that becomes unavoidable when one asks whether a named causal edge is intrinsic: the edge belongs to a schedule presentation, not to the underlying lift alone.

Our terminology is deliberately local to this normalization problem. Event structures and higher-dimensional models provide general formalisms for causality and concurrency [3, 4]; here the causal hypergraph is instead the finite obstruction hypergraph of optional normalization sites. The pushdown background is standard [1], but the classification below uses the arithmetic of one particular shortest-completion family rather than a general concurrency semantics.

Main results. The arbitrary-coprime classification consists of four ingredients.

1. A uniform compiler maps every local observation to the unique shortest invariant-preserving direct center return.
2. The complete baseline-portable site universe has at most eight vertices for $\min(p, q) \geq 2$, with explicit parity-dependent threshold inequalities.
3. The exact minimal fatal supports are determined in the regimes $\min(p, q) \geq 3$, $\min(p, q) = 2$, and $\min(p, q) = 1$.
4. The resulting admissible family is always downward closed for this canonical weighted-balance schedule problem, but downward closure fails in the abstract General-Theory setting.

2 From the original 2:1 machine to schedule geometry

The present classification was not obtained by beginning with an abstract hypergraph. Its starting point is the author's original completion-aware 2:1 DPDA, whose center/left/right organization is shown in fig. 1. In that machine, ordinary binary transitions perform stack cancellation and replacement, while the end-of-input completion rules convert a bounded side-state offset into explicit completion demand. The side states therefore carry part of the same arithmetic deficit that is represented by the stack; they are not independent counters.

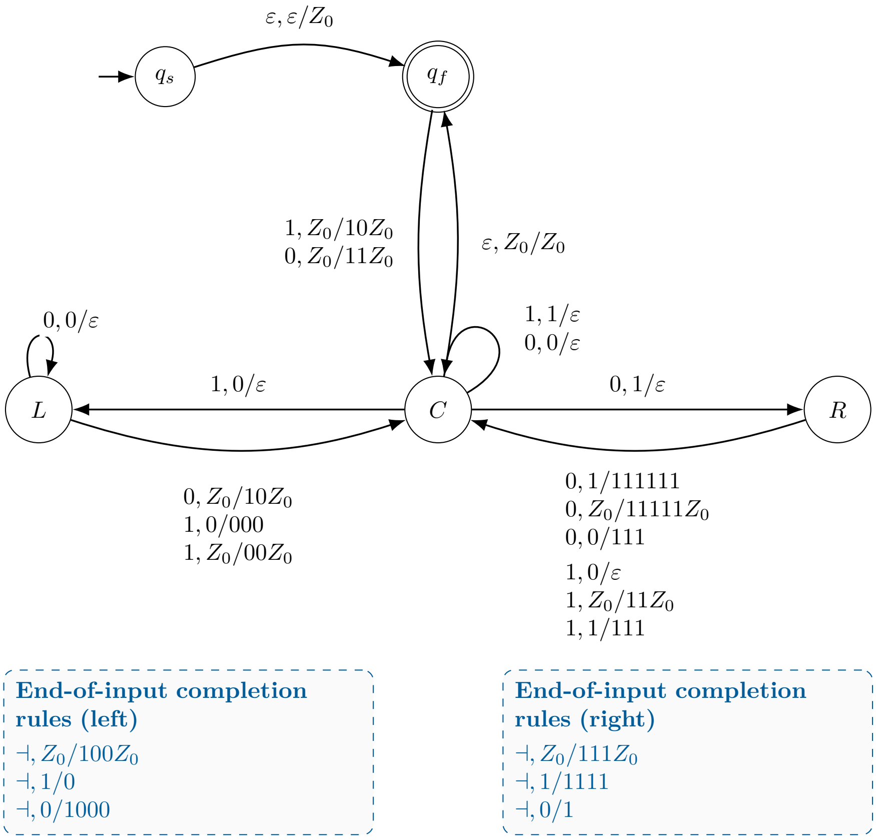

Figure 1: The original completion-aware 2:1 DPDA, in the empty-word-included formal variant. Solid arcs process binary input; dashed callout boxes list only end-of-input completion rules. The machine is the historical source of the optional-normalization questions classified here, but the arbitrary-coprime proofs do not use this transition table as a hypothesis.

The later $p{:}q$ normalization program isolates this observation as a residual/shortest-demand invariant and replaces the accidental constants of the 2:1 table by a centered family of debt states. Once that family is written in literal-demand form, some local returns to the center are mandatory and others may be performed eagerly without changing the intended semantics. Those optional eager returns are exactly the objects that become *schedule sites* in this paper.

Thus the research path is

$$\boxed{\begin{array}{c}\text{original 2:1 completion mechanism} \longrightarrow \text{centered } p{:}q \text{ literal demand} \\ \longrightarrow \text{optional eager sites} \longrightarrow \text{causal hypergraph}\end{array}}$$

The figure is included to make this provenance explicit. None of the arbitrary-coprime classification theorems below assumes the original transition table as a hypothesis; the proofs use the centered $p{:}q$ compiler, reset envelope, and portability conditions developed from its completion semantics.

3 Centered literal-demand normalization and schedule presentations

Put

$$m = p + q, \qquad h = \left\lfloor \frac{m-1}{2} \right\rfloor, \qquad K = (q, p).$$

The centered literal-demand lift has finite-control debt states

$$D_{-h}, D_{-h+1}, \dots, D_h$$

and a center state C. The semantic residual is represented by a canonical shortest nonnegative demand vector

$$\mu(r) = \arg\min\{x + y : x, y \geq 0,\ px - qy = r\},$$

and the chosen literal stack order is denoted by $\eta(x, y)$.

For debt state D_d define the virtual demand

$$V_d = \mu(dm).$$

A local work-top observation is a triple

$$o = (d, a, z), \qquad |d| \leq h, \quad a, z \in \{0, 1\},$$

meaning that the current state is D_d, the next input letter is a, and the visible stack symbol is z.

Definition 3.1 (Schedule presentation). *A schedule presentation consists of a fixed mandatory centered baseline policy together with a finite optional-site menu and, for each site, a designated eager-return rule. A schedule is a subset of the menu; enabled optional rules may pre-empt the corresponding baseline transition.*

Definition 3.2 (Admissibility and fatal supports). *A schedule H is* admissible *if the resulting deterministic machine recognizes exactly $L_{p,q}$. A support F is* fatal *if every schedule containing F is inadmissible, and is* minimal fatal *if no proper subset is fatal.*

The distinction between a lift and a schedule presentation will be essential in Section 10: the physical transition system does not determine which optional sites are named or offered.

4 The uniform eager-return compiler

Let $e_0 = (1, 0)$ and $e_1 = (0, 1)$. For an observation $o = (d, a, z)$ define

$$B_o = e_z - e_a + V_d$$

and

$$\kappa(o) = \max\left\{ \left\lceil -\frac{(B_o)_0}{q} \right\rceil, \left\lceil -\frac{(B_o)_1}{p} \right\rceil \right\}.$$

Set

$$R_o = B_o + \kappa(o)K.$$

The associated local eager rule is

$$E_o : (D_d, zT) \xrightarrow{a} (C, \eta(R_o)T).$$

Theorem 4.1 (Uniform centered eager-return compiler). *For every local observation o, the vector R_o is the unique shortest nonnegative replacement vector that preserves the residual invariant and returns directly to the center. Hence E_o is the canonical direct eager-return candidate at o.*

Proof. Any direct return to the center must compensate the change in the visible stack cell and the consumed input, hence its replacement vector lies in the affine lattice

$$B_o + \mathbb{Z}K.$$

The only admissible replacements are nonnegative vectors. Adding one copy of $K = (q, p)$ increases both coordinates, so the smallest integer carry making both coordinates nonnegative is exactly $\kappa(o)$. Any smaller carry has a negative coordinate, while any larger carry increases total length by $p + q$. Therefore R_o is feasible and uniquely length-minimal. □

This formula recovers, for example, the six explicit $3:2$ eager words used in the finite stress test. The important point here is not those special values but the existence of one ratio-uniform compiler from which the whole site universe can be derived.

4.1 The mandatory reset envelope

Write

$$\mu((h+1)m) = (x_+, y_+), \qquad \mu(-(h+1)m) = (x_-, y_-).$$

Define

$$\mathcal{R}^0_{p,q} = \{\mu(jm+q) : -h-1 \le j \le h\} \cup \{(x_+ + n, y_+) : n \ge 0\} \cup \{(x_-, y_- + n) : n \ge 0\}.$$

Lemma 4.2 (Mandatory-WRITE reset envelope)**.** *Every mandatory WRITE in the centered $p:q$ baseline lands at a canonical center pair belonging to $\mathcal{R}^0_{p,q}$.*

Proof. A bottom WRITE from D_d on input 1 produces $\mu(dm+q)$, whereas input 0 produces

$$\mu(dm-p) = \mu((d-1)m+q),$$

which gives the finite seed range $-h-1 \le j \le h$. At the positive outer boundary, the untouched suffix is a unary 0-tail, so the reset ray is $(x_+ + n, y_+)$. The negative side is symmetric. □

Definition 4.3 (Baseline-portable site)**.** *A baseline episode starts at a center configuration $(C, \eta(x,y))$ with $(x,y) \in \mathcal{R}^0_{p,q}$, follows only mandatory MATCH/SEARCH transitions, and stops before the next mandatory bottom or outer WRITE. A compiler site is* baseline-portable *if its eager rule lands on the canonical center representative for every source exposed in every baseline episode.*

Baseline portability is deliberately weaker than singleton admissibility: an eager reset can be safe on all baseline sources and later become unsafe on a source created by its own earlier activation. This self-exposure mechanism is responsible for the extra singleton-fatal vertices below.

5 The finite baseline-portable site universe

Assume first that $p \ge q \ge 2$. For $j \ge 0$, the bottom-seed formula is

$$\mu(jm+q) = (j+q, p-j-1).$$

For $j = -t < 0$, write $t = kq + s$, $0 \le s < q$; then

$$\mu(-tm+q) = \begin{cases} (0, km-1), & s = 0, \\ (q-s, p+km+s-1), & 1 \le s < q. \end{cases}$$

For $1 \le d \le h$,

$$V_d = (d+q, p-d).$$

The three nontrivial positive-debt compiler count vectors are

$$M_d = (d+q, p-d), \qquad P_d = (d+q+1, p-d-1), \qquad N_d = (d+q-1, p-d+1),$$

with imbalances

$$\Delta(M_d) = 2d + q - p,$$

$$\Delta(P_d) = 2d + q - p + 2,$$

$$\Delta(N_d) = 2d + q - p - 2.$$

Negative-debt compiler words are one-heavy except for the trivial forced return that is already part of the baseline.

A central reset word excludes almost every interior observation. If m is odd, the diagnostic seed is (h, h) with word $(10)^h$; if m is even, it is $(h, h+1)$ with word $(10)^h1$. Tracking untouched suffixes shows that all negative-interior candidates fail portability and leaves at most two positive-interior survivors.

Theorem 5.1 (Eight-Site Theorem). *Let $p \geq q \geq 2$ be coprime and let $h = \lfloor (p+q-1)/2 \rfloor$.*

If $p+q$ is odd, the baseline-portable site universe consists exactly of the six core sites

$$(-h, 0, 0),\ (-h, 1, 0),\ (-h, 1, 1),\ (h-1, 0, 0),\ (h-1, 1, 0),\ (h, 0, 0),$$

together with

$$X = (h-q+1, 0, 1) \quad \Longleftrightarrow \quad p > 5q-5,$$

and

$$Y = (h-q, 1, 1) \quad \Longleftrightarrow \quad p > 5q-1.$$

If $p+q$ is even, the universe consists exactly of the four core sites

$$(-h, 1, 1),\ (h, 0, 0),\ (h, 0, 1),\ (h, 1, 1),$$

together with

$$X = (h-q+1, 0, 0) \quad \Longleftrightarrow \quad p > 5q-5,$$

and

$$Y = (h-q, 1, 0) \quad \Longleftrightarrow \quad p > 5q.$$

In particular,

$$|\mathcal{S}^{\mathrm{bp}}_{p,q}| \leq 8.$$

For $q > p \geq 2$, the complete list is obtained by the physical duality

$$(p, q, d, a, z) \longleftrightarrow (q, p, -d, 1-a, 1-z).$$

Proof. The proof is an exact arithmetic elimination. The central diagnostic removes every negative-interior observation and all but two positive-interior observations; exact boundary fibers give the core; the two survivors are decided by primitive-kernel minimality. All formulas, including the negative-seed decomposition and the parity-specific threshold simplifications, are given in Appendix A. □

6 The high-minimum regime: every core schedule is safe

Let $\mathrm{Core}_{p,q}$ denote the parity-dependent core in Theorem 5.1. For a named site A, write $\mathcal{E}_A$ for the set of visible source stack words that can occur at that site during an episode under the chosen core schedule. These source envelopes are physical reachability sets, not new semantic states.

Theorem 6.1 (Core downward closure). *If $\min(p, q) \geq 3$, then every subset $H \subseteq \mathrm{Core}_{p,q}$ is admissible. Equivalently,*

$$2^{\mathrm{Core}_{p,q}} \subseteq \mathrm{Adm}_{p,q}\,.$$

No minimal fatal support is contained entirely in the core.

Proof. By duality assume $p \geq q \geq 3$. Induct over reset episodes. Every enabled core activation returns to a canonical minimum-demand word. It therefore suffices to show that the source fibers of core sites stay within protected unary envelopes after any sequence of core activations.

If $m = 2h + 1$, label the six core sites

$$A = (-h, 0, 0),\ B = (-h, 1, 0),\ C = (-h, 1, 1),$$

$$D = (h - 1, 0, 0),\ E = (h - 1, 1, 0),\ F = (h, 0, 0).$$

Uniformly over all core schedules,

$$\mathcal{E}_A = \mathcal{E}_B = \{0\}, \qquad \mathcal{E}_C \subseteq 1^+,$$

and

$$\mathcal{E}_D, \mathcal{E}_E, \mathcal{E}_F \subseteq 0^+.$$

Negative-core resets contain fewer than $q \leq h$ zeros, so they cannot create a new negative-boundary top-0 source. Positive-core outputs have one-count at most $h - 1$, so their mixed part is exhausted before the next positive-boundary source. The eager compiler has the same tail orientation as the protected source, and kernel minimality is preserved.

If $m = 2h + 2$, the four core sites satisfy

$$\mathcal{E}_A \subseteq 1^+, \qquad \mathcal{E}_B \subseteq 0^+, \qquad \mathcal{E}_C = \mathcal{E}_D = \{1\}.$$

The same minority-count argument prevents a negative reset from generating a new positive-boundary source and prevents a positive reset from generating an additional top-1 source at D_h. Hence every optional activation is canonical and kernel-reduced. The residual invariant and the baseline terminal argument then imply correctness. $\square$

7 Extra-site self-exposure for $\min(p, q) \geq 3$

The extra vertices X, Y are baseline-portable precisely because their first activation stays below the primitive kernel threshold. Repeating the same activation increases a hidden alternating prefix until one full kernel packet becomes trapped.

For odd $m = 2h + 1$, define

$$S_k^X = (10)^{k(q-1)} 1^{2q-2}, \qquad S_k^Y = (10)^{kq} 1^{2q}.$$

For even $m = 2h + 2$, define

$$S_k^X = 0(10)^{k(q-1)} 1^{2q-3}, \qquad S_k^Y = 0(10)^{kq} 1^{2q-1}.$$

The first source S_0 is reached by an explicit monotone baseline prefix. Between consecutive X-activations one reads $1^{p-q+1}0$; between consecutive Y-activations one reads 1^{p-q}. Thus the alternating exponent grows by $q - 1$ or q respectively.

Lemma 7.1 (One-kernel trapping)**.** *Let*

$$k_X = \left\lceil \frac{p - (h + 2q - 2)}{q - 1} \right\rceil, \qquad k_Y = \left\lceil \frac{p - (h + 2q)}{q} \right\rceil.$$

At the first nonminimal repeated activation, the physical target differs from the canonical minimum demand by exactly one primitive kernel packet $K = (q, p)$.

Proof. At the fatal activation the one-count is $p + e$ with $0 \leq e \leq q - 1$ (and $e \leq q - 2$ in the X case), while the zero-count is at least q. Subtracting one K therefore remains nonnegative, while subtracting two cannot. Thus the target is exactly $\mu(r) + K$. $\square$

Feeding the shortest semantic completion of r removes the minimal component and leaves exactly one balanced kernel packet physically present. The final semantic residual is zero but the machine is not in the balanced accepting configuration.

Theorem 7.2 (Complete schedule classification for $\min(p,q) \geq 3$)**.** *For every coprime p, q with $\min(p,q) \geq 3$,*

$$\boxed{\mathrm{Adm}_{p,q} = 2^{\mathrm{Core}_{p,q}}.}$$

Every present extra site is an upward-fatal singleton conflict, and these are the only minimal fatal supports.

Proof. Every core subset is admissible by Theorem 6.1. The Y self-exposure witness meets no other optional site, hence every schedule containing Y is bad. The X witness meets no core site and at most the neighboring Y observation; if Y is disabled it proves X fatal, while if Y is enabled the Y witness already proves failure. Thus every schedule containing either extra is inadmissible. □

8 Low-minimum regimes and the arbitrary-coprime theorem

8.1 The line $\min(p,q) = 2$

By duality it suffices to take $q = 2$, so $p \geq 3$ is odd. The six-site core has exactly two minimal fatal supports,

$$\{A, D\}, \qquad \{B, D\}.$$

The two possible extra sites are

$$\begin{aligned} X_1 &= (h-1, 0, 1) \quad (p \geq 7), \\ X_2 &= (h-2, 1, 1) \quad (p \geq 11). \end{aligned}$$

and both are singleton-fatal.

Theorem 8.1 (Exact $\min(p,q) = 2$ classification)**.** *For $q = 2$,*

$$\mathrm{Adm}_{p,2} = \{H \subseteq \mathcal{S}^{\mathrm{bp}}_{p,2} : X_1, X_2 \notin H,\ \{A, D\} \not\subseteq H,\ \{B, D\} \not\subseteq H\},$$

with absent extras omitted. Exactly 40 *schedules are admissible.*

Proof. Appendix B.1 gives a self-contained reset-closure proof of sufficiency and explicit robust witnesses for the two fork pairs $\{A, D\}$ and $\{B, D\}$; the six-site core therefore has exactly 40 safe schedules. Each present extra site is an upward-fatal singleton by the one-kernel trapping certificate, so every admissible expanded schedule excludes all extras. Conversely, after the extras are excluded the machine is exactly one of the safe core schedules. Thus the admissible count remains 40 independently of how many extras are present. □

8.2 The exceptional line $q = 1$

For $q = 1$, negative minimum demands are unary, so the negative corridor has no order freedom. The complete baseline-portable site universe is exceptional.

For $p = 1$ it is empty. For $p = 2$ the four sites are

$$X = (0, 1, 0), \quad A = (-1, 1, 1), \quad B = (-1, 0, 0), \quad G = (1, 0, 0).$$

For odd $p = 2h + 1 \geq 3$, the always-present sites are

$$A = (-h, 1, 1), \qquad D = (h, 1, 1), \qquad E = (h, 0, 0),$$

and for $p \geq 5$ also

$$B = (h-1, 1, 0), \qquad C = (h, 0, 1).$$

For even $p = 2h \geq 4$, the sites

$$A = (-h, 1, 1),\ B = (-h, 0, 0),\ C = (-h, 1, 0),$$
$$E = (h-1, 0, 0),\ F = (h-1, 1, 0),\ G = (h, 0, 0)$$

are always present, and $D = (h-1, 1, 1)$ appears for $p \geq 6$.

Theorem 8.2 (Exact $q = 1$ schedules)**.** *Up to physical duality:*

1. $(p, q) = (1, 1)$*: the site universe is empty and there is one admissible schedule.*
2. $(2, 1)$*: the minimal fatal supports are* $\{B, X\}$ *and* $\{B, G\}$*; exactly* 10 *schedules are admissible.*
3. *odd* $p \geq 3$*:*
$$\mathrm{Adm}_{p,1} = \{\varnothing, \{A\}\}.$$
4. *even* $p \geq 4$*: the maximal safe sectors are*
$$\mathcal{N} = \{A, B, C\}, \qquad \mathcal{P} = \{A, F, G\},$$
while D *and* E *are singleton-fatal when present and the cross-pairs*
$$\{B, F\}, \{B, G\}, \{C, F\}, \{C, G\}$$
are fatal. Exactly 14 *schedules are admissible.*

Proof. The site-universe elimination and the balanced fatal certificates are recorded in Appendix B. The safe cases are proved by episode-closed reset families: for odd p, only the unary negative ray generated by A is added; for even p, every safe schedule lies in one of the two closed sectors $2^{\mathcal{N}}$ or $2^{\mathcal{P}}$. The listed singleton and cross-pair witnesses trap a primitive kernel packet at residual zero. The small case $(2, 1)$ is verified by its four-site transition geometry, with explicit witnesses 111001 and 111110001. □

Theorem 8.3 (Arbitrary-coprime causal-hypergraph classification)**.** *For every coprime positive pair* (p, q)*, the canonical baseline-portable weighted-balance schedule family has an explicit finite set of minimal fatal supports given by Theorems 7.2, the* min $= 2$ *theorem, and Theorem 8.2, together with physical duality. Consequently*

$$H \in \mathrm{Adm}_{p,q}, \quad H' \subseteq H \quad \Longrightarrow \quad H' \in \mathrm{Adm}_{p,q}\,.$$

Equivalently, $\mathrm{Adm}_{p,q}$ *is the independence complex of the exact fatal-support hypergraph, in the standard simplicial-complex sense [2].*

9 Downward closure is not universal

Theorem 8.3 is a theorem about the canonical weighted-balance family, not a theorem about arbitrary exact lifts. The distinction is real.

Consider one semantic state m with two semantic letters x, y, both acting as the identity. Let the physical states be c, d, b, all projecting to m. Assign terminal values

$$\alpha(c) = \alpha(d) = 0, \qquad \alpha(b) = 1.$$

The baseline transitions are

$$c \xrightarrow{x} c, \quad c \xrightarrow{y} c,$$
$$d \xrightarrow{x} c, \quad d \xrightarrow{y} b,$$

with b absorbing. Add two optional sites:

$$A : c \xrightarrow{x} d, \qquad B : d \xrightarrow{y} c.$$

Theorem 9.1 (Two-site failure of universal downward closure)**.** *For this exact schedule environment,*

$$\mathrm{Adm} = \{\varnothing, \{B\}, \{A, B\}\}.$$

In particular $\{A, B\}$ is admissible but $\{A\}$ is not.

Proof. The empty schedule and $\{B\}$ never leave c. Under $\{A\}$, the word xy follows $c \xrightarrow{x} d \xrightarrow{y} b$ and fails. Under $\{A, B\}$, the same first step reaches d but the enabled B rule repairs the second step and returns to c. Thus adding an optional rule can repair a failure created by another optional rule. □

This two-site example is minimal among counterexamples with an admissible empty schedule: with only one optional site, the only possible failure of downward closure would require the empty schedule to be bad and the singleton schedule good.

10 Causal persistence depends on the presentation category

A named edge such as $A \to B$ refers to two designated optional sites and to a first-hit or pre-emption relation between them. This data is not contained in the underlying lift alone.

Proposition 10.1 (Typing of named causal edges)**.** *A fixed exact lift does not determine a named causal edge. The optional-site menu and the local alternative rules are part of the schedule presentation, so forgetting the presentation can change the causal graph without changing the baseline lift.*

For a named edge e, let $\Pi(e) = (\pi_0, \pi_1, \pi_2)$ record persistence under (i) site-preserving coordinate isomorphism, (ii) a specified fixed-literal-policy presentation class, and (iii) broad exact physical re-coordination. Because the middle coordinate depends on what presentation data are frozen, its class is written as a subscript.

For the classical $2:1$ literal one-top policy, consider two presentation categories. In the first, the baseline policy is fixed but the optional-site menu may vary. In the second, the designated A, B cells and the finite causal certificate are preserved.

Theorem 10.2 (Exact $A \to B$ category bifurcation)**.** *For the classical $2:1$ baseline literal one-top policy,*

$$\Pi_{\mathsf{flex}}(A \to B) = (1, 0, 0)$$

in the site-flexible fixed-policy class, while

$$\Pi_{\mathsf{cert}}(A \to B) = (1, 1, 0)$$

in the certificate/site-preserving class. Hence the middle persistence bit has no category-independent value.

Proof. The same all-lazy baseline lift admits two presentations. With menu $\{A, B, C\}$ the classical first-hit certificate yields $A \to B$; with menu $\{A, C\}$ the named B site is absent and the directed edge cannot be formed. Hence the middle bit is 0 in the site-flexible class. In the certificate-preserving class the cells

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

and the distinguishing continuation 1 are part of the morphism data; the same noncanonical first-hit fiber therefore survives every allowed change. Broad re-coordination can remove the named phenomenon. Appendix C gives the typed formulation. □

11 Synthesis

The rational weighted-balance family has a particularly rigid schedule geometry: its optional-site universe is uniformly finite, its fatal supports are explicit, and its admissible schedules form a simplicial independence complex for every coprime ratio. The proof depends on arithmetic facts specific to shortest-demand normalization: canonical reset envelopes, minority bounds, primitive-kernel reduction, and self-exposure recurrences.

Two boundaries are equally important. First, downward closure is not a universal law of exact-lift scheduling. Second, a named causal edge is not a property of a lift without a presentation category. Together these results separate three levels that should not be conflated: exact family-specific schedule classification, abstract schedule axioms, and persistence under representation change.

Acknowledgment of scope

The paper deliberately makes no claim that all exact pushdown schedule systems are independence complexes, and no claim that causal signatures are invariant under unrestricted re-coordination of the physical lift. The classification is exact for the canonical baseline-portable weighted-balance schedule presentation described above.

A Arithmetic details for the Eight-Site Theorem

This appendix records the elimination argument behind Theorem 5.1; it is included so that the arbitrary-coprime vertex classification can be checked without relying on a finite search.

Put $m = p + q$ and $h = \lfloor (m-1)/2 \rfloor$, and assume $p \geq q \geq 2$. For the bottom-WRITE seed family $B_j = \mu(jm + q)$ one has, for $j \geq 0$,

$$B_j = (j + q, p - j - 1).$$

For $j = -t < 0$, write $t = kq + s$, $0 \leq s < q$. Then

$$B_{-t} = \begin{cases} (0, km - 1), & s = 0, \\ (q - s, p + km + s - 1), & 1 \leq s < q. \end{cases}$$

For $1 \leq d \leq h$,

$$V_d = \mu(dm) = (d + q, p - d).$$

Thus the positive-debt compiler count vectors are

$$M_d = (d + q, p - d), \quad P_d = (d + q + 1, p - d - 1), \quad N_d = (d + q - 1, p - d + 1),$$

with imbalances

$$2d + q - p, \qquad 2d + q - p + 2, \qquad 2d + q - p - 2.$$

Every nontrivial negative-debt compiler word is one-heavy; the only exception is the forced $d = -1$, input/top 1/0 return already contained in the baseline.

Lemma A.1 (Central diagnostic exclusion)**.** *For $p \geq q \geq 2$, no negative-interior observation is baseline-portable. If m is odd, the only possible positive-interior survivors are*

$$(h - q + 1, 0, 1), \qquad (h - q, 1, 1),$$

and if m is even the only possible positive-interior survivors are

$$(h - q + 1, 0, 0), \qquad (h - q, 1, 0).$$

Proof. If $m = 2h + 1$, the seed $B_{h-q} = (h, h)$ has canonical word $(10)^h$. At an interior debt d, the untouched suffix is $(10)^{h-d-1}$ under a visible top 0 and $0(10)^{h-d-1}$ under a visible top 1. A top-0 eager concatenation is canonical only when its compiler word is balanced; all relevant interior imbalances are odd. A top-1 concatenation is canonical only when the compiler has exactly one excess 1, which gives precisely the two displayed equations for d.

If $m = 2h + 2$, the central seed is $(h, h + 1)$ with word $(10)^h1$. The corresponding suffixes are $(10)^{h-d-1}1$ and $0(10)^{h-d-1}1$. A top-0 candidate must have zero imbalance, while the parity of the positive-debt compiler imbalances excludes every top-1 candidate. Solving the zero-imbalance equations gives the two even-parity survivors. For negative debt, the compiler has a terminal 1-tail whereas the central diagnostic exposes an untouched 0 beyond the visible cell; the eager target is therefore not in canonical alternating-tail form. This excludes all negative-interior sites. □

Lemma A.2 (Boundary fibers)**.** *The always-present boundary sites are exactly the parity-dependent cores in Theorem 5.1.*

Proof. For odd m, the negative boundary has a singleton top-0 source and unary top-1 sources; at $d = h - 1$ every top-0 source is unary, and at $d = h$ every source is unary top-0. A source 11 at $d = h - 1$ eliminates the remaining top-1 candidates. For even m, the negative boundary has unary top-1 sources but also the diagnostic source 00, while at $d = h$ the top-0 sources are unary and the only top-1 source is 1; the central sources 01 and 101 eliminate the remaining $d = h - 1$ candidates. Direct substitution into the compiler gives exactly the core lists stated in Theorem 5.1. □

For the two odd-parity survivors the common compiler word is $(10)^h1$. Their exposed source one-counts are bounded by $2q - 2$ and $2q$, with equality attained by baseline sources. Kernel reduction is therefore equivalent to

$$h + 2q - 2 < p, \qquad h + 2q < p,$$

which simplify to $p > 5q - 5$ and $p > 5q - 1$. For even m the common compiler is $(10)^{h+1}$, the extremal untouched one-counts are $2q - 3$ and $2q - 1$, and the same calculation gives $p > 5q - 5$ and, using coprimality of odd p, q, $p > 5q$. This completes the proof of Theorem 5.1.

B Low-minimum certificates

The low-minimum cases are included explicitly because they are the only places where non-singleton fatal supports survive.

B.1 Self-contained odd-p:2 fork certificate

Fix $q = 2$ and write

$$p = 2h - 1, \qquad m = p + q = 2h + 1, \qquad h \geq 2.$$

For $t \geq 1$ put

$$\epsilon_t = t \bmod 2, \qquad k_t = \left\lceil \frac{t}{2} \right\rceil, \qquad \beta_t = t + pk_t, \qquad \Gamma_t = \eta(\epsilon_t, \beta_t).$$

The six core rules are, in literal stack notation,

$$\begin{array}{r|l}
A : (D_{-h}, 0T) \xrightarrow{0} (C, \Gamma_h T) & B : (D_{-h}, 0T) \xrightarrow{1} (C, \Gamma_{h-1} T) \\
C : (D_{-h}, 1T) \xrightarrow{1} (C, \Gamma_h T) & D : (D_{h-1}, 0T) \xrightarrow{0} (C, (10)^h 0T) \\
E : (D_{h-1}, 0T) \xrightarrow{1} (C, (10)^{h-1}000T) & F : (D_h, 0T) \xrightarrow{0} (C, (10)^{h-1}000T).
\end{array}$$

The arithmetic identities needed below are

$$\mu(-tm) = (\epsilon_t, \beta_t), \qquad \mu(jm+2) = (j+2, p-j-1) \quad (0 \le j \le h),$$

$$\mu(-tm+2) = (\epsilon_t, \beta_t - 1) \quad (1 \le t \le h+1), \qquad \mu((h+1)m) = (h+3, h-2).$$

They follow by direct substitution in $px - 2y = r$ and kernel reduction by $(2, p)$.

Define the canonical baseline reset family

$$\mathcal{R}_h^{\mathrm{base}} = \{\mu(jm+2) : -h-1 \le j \le h\} \cup \{(x, h-2) : x \ge h+3\} \cup \{(\epsilon_{h+1}, y) : y \ge \beta_{h+1}\}.$$

Two episode-closed repertoires separate the safe schedule regimes:

$$\mathcal{R}_{\neg D,h} = \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \cup \{(\epsilon_{h-1}, \beta_{h-1})\} \cup \{(x, h-1) : x \ge h+2\},$$
$$\mathcal{R}_{D,h} = \mathcal{R}_h^{\mathrm{base}} \cup \{(\epsilon_h, y) : y \ge \beta_h\} \cup \{(x, h-1) : x \ge h+2\} \cup \{(x, h) : x \ge h+1\}.$$

Every pair in either repertoire has minority coordinate at most h. Consequently an episode beginning in one of these repertoires has the following exact boundary-source geometry: in $\mathcal{R}_{\neg D,h}$, a D_{-h} source with visible 0 has remaining stack exactly 0; in $\mathcal{R}_{D,h}$ it has remaining stack 0^n, $n \ge 1$, with $n \ge 2$ arising only from the D-ray. In either repertoire, D_{-h} with visible 1, D_{h-1} with visible 0, and D_h with visible 0 have unary remaining stacks. This follows because MATCH and interior SEARCH only delete the visible top cell, so every pre-reset stack is a suffix of a canonical reset word; an opposite symbol beyond the displayed unary suffix would force minority size at least $h+1$.

For a unary source of length $n \ge 1$, the six eager targets have canonical count pairs

$$\begin{array}{r|c} A:0 & (\epsilon_h, \beta_h) \\ B:0 & (\epsilon_{h-1}, \beta_{h-1}) \\ C:1^n & (\epsilon_h, \beta_h + n - 1) \\ D:0^n & (n+h, h) \\ E:0^n & (n+h+1, h-1) \\ F:0^n & (n+h+1, h-1). \end{array}$$

Hence baseline resets together with any enabled subset of A, B, C, E, F preserve $\mathcal{R}_{\neg D,h}$ when D is disabled; when D is enabled but A, B are disabled, baseline resets together with C, D, E, F preserve $\mathcal{R}_{D,h}$. Therefore all $2^5 = 32$ schedules with $D = 0$ are safe, and with $D = 1$ precisely the $2^3 = 8$ schedules having $A = B = 0$ are safe.

It remains to prove necessity. For every $x \ge h+1$,

$$(C, \eta(x,h)) \xrightarrow{0^{2h-1}} (D_{-h}, 0^{x-h+1}),$$

so enabling D expands the common A/B source fiber from the singleton $\{0\}$ to the entire unary ray $\{0^n : n \ge 1\}$. There is a uniform prefix u_h reaching $(D_{-h}, 000)$ and meeting no optional site except D. Put

$$v_h = \begin{cases} 1^{(h-1)m/2}, & h \text{ odd}, \\ 0\,1^{2h+(h-2)m/2}, & h \text{ even}, \end{cases} \qquad u_h = v_h 1^{2h-1} 0^{2h}.$$

Then u_h has the stated property: the increment block from an empty stack advances the centered debt by two, the middle block reaches $(D_{h-1}, 00)$, the next zero invokes D, and the final zeros expose $(D_{-h}, 000)$ without meeting A, B, C, E, F earlier.

Finally, Γ_t has a suffix 1^p for every $t \ge 1$; write $\Gamma_t = z_t 1^p$. At the common source $(D_{-h}, 000)$,

$$A: \ (D_{-h}, 000) \xrightarrow{0} (C, z_h 1^p 00), \qquad B: \ (D_{-h}, 000) \xrightarrow{1} (C, z_{h-1} 1^p 00).$$

Reading z_h or z_{h-1} by MATCHes leaves $(C, 1^p00)$. This word contains p ones and two zeros, so its residual is zero, but it is a nonempty primitive kernel packet and is therefore nonaccepting. The complete fatal witnesses

$$w_A^{(h)} = u_h 0 z_h, \qquad w_B^{(h)} = u_h 1 z_{h-1}$$

show that $\{A, D\}$ and $\{B, D\}$ are robust fatal supports. Thus the six-site core has exactly 40 admissible schedules and no other minimal fatal support.

The extra baseline-portable sites

$$X_1 = (h-1, 0, 1) \quad (p \geq 7), \qquad X_2 = (h-2, 1, 1) \quad (p \geq 11)$$

are individually upward-fatal by the one-kernel trapping mechanism of Section 7. Excluding them reduces the expanded problem to the proved six-site fork classification, so the admissible count remains exactly 40.

For $q = 1$ the full vertex sets are as follows. At $(2, 1)$,

$$\mathcal{S}_{2,1}^{\mathrm{bp}} = \{X = (0,1,0),\ A = (-1,1,1),\ B = (-1,0,0),\ G = (1,0,0)\}.$$

For odd $p = 2h+1 \geq 3$, $A = (-h,1,1), D = (h,1,1), E = (h,0,0)$ are always present and $B = (h-1,1,0), C = (h,0,1)$ occur for $p \geq 5$. For even $p = 2h \geq 4$, the always-present sites are

$$A = (-h,1,1),\ B = (-h,0,0),\ C = (-h,1,0),\ E = (h-1,0,0),\ F = (h-1,1,0),\ G = (h,0,0),$$

and $D = (h-1,1,1)$ is present iff $p \geq 6$.

Lemma B.1 (Explicit $q = 1$ fatal certificates)**.** *The fatal supports in Theorem 8.2 admit balanced witnesses that end at residual zero in a nonaccepting physical configuration.*

Proof. For odd $p = 2h+1$, the positive sites are witnessed by

$$w_B = 1^{p(p-2)}0^{p-2},$$

$$w_C = 1^{2h(h+1)-1}(0\,1^{p-1})^{h-1}0^h,$$

$$w_D = 1^{2h(h+1)}0^p 1^{2h(h+1)+1}, \qquad w_E = 1^{2h^2+4h+1}0^{p+1}1^{2h(h+1)+1}.$$

The B-witness meets no other optional site; with B disabled the C- and D-witnesses are unpreempted, and with B, D disabled the E-witness is unpreempted. Site A acts only on the unary negative corridor and preserves an episode-closed unary reset family. Hence only $\varnothing$ and $\{A\}$ are admissible.

For even $p = 2h$, the two safe sectors are $\mathcal{N} = \{A, B, C\}$ and $\mathcal{P} = \{A, F, G\}$. The individually fatal sites have witnesses

$$w_D = 1^{p(p-3)}0^{p-3} \quad (h \geq 3),$$

$$w_E = 1^{(h+1)(p-1)}(0\,1^{p-1})^{h-1}0^h.$$

The four cross-pair certificates are

$$w_{BF} = 1^{h(p+1)}0^p 1^{h(p-1)}, \qquad w_{CF} = 1^{h(p+1)}0^{p-1}1^{h(p-3)},$$

$$w_{BG} = 1^{h(p+3)}0^{p+1}1^{h(p-1)}, \quad w_{CG} = 1^{h(p+3)}0^p 1^{h(p-3)}.$$

They trap exactly one primitive kernel packet after the indicated mixed pair has been activated. Finally, for $(p, q) = (2, 1)$ the two balanced words 111001 and 111110001 witness $\{B, X\}$ and $\{B, G\}$ and both end at $(C, 110)$, a nonaccepting ordering of the primitive kernel packet. These certificates, together with the episode-closed safe sectors, prove the stated low-minimum classification. $\square$

C Persistence signature and category typing

A schedule presentation is a triple

$$\mathfrak{P} = (\mathcal{L}_0, E, \{\delta^e\}_{e \in E}),$$

where $\mathcal{L}_0$ is the fixed baseline lift, E is the named optional-site menu, and each e carries its alternative local rule. A named directed edge therefore belongs to $\mathfrak{P}$, not to $\mathcal{L}_0$ alone.

For a named edge e define its three-level persistence signature

$$\Pi(e) = (\pi_0, \pi_1, \pi_2) \in \{0,1\}^3,$$

where π_0 records persistence under site-preserving coordinate isomorphism, π_1 records persistence in a specified fixed-literal-policy presentation class, and π_2 records persistence under broad exact physical re-coordination. The middle class must therefore be named as part of the datum.

For the classical 2:1 edge $A \to B$, the all-lazy baseline admits both the menu $\{A, B, C\}$, where the directed dependency is present, and the menu $\{A, C\}$, where it is absent. This proves the middle bit is 0 in the site-flexible class. If instead the designated A, B cells and the finite first-hit certificate

$$1111100 \rightsquigarrow (R, 00), \qquad (R, 00) \xrightarrow[B]{0} (C, 1110)$$

with distinguishing continuation 1 are preserved, the same unsafe first-hit fiber survives every allowed presentation change; the middle bit is then 1. Broad re-coordination can eliminate the named phenomenon. Thus

$$\Pi_{\mathrm{flex}}(A \to B) = (1, 0, 0), \qquad \Pi_{\mathrm{cert}}(A \to B) = (1, 1, 0).$$

Statements and Declarations

Author contributions. Alp Eren Bütün is the sole author and was responsible for conceptualization, methodology, formal analysis, investigation, theorem development, computational validation where reported, visualization, writing–original draft, and writing–review and editing.

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Data and code availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, formulas, finite certificates, and proofs required for the mathematical results are contained in the manuscript. Any computations reported in the manuscript are independent regression checks and are not proof premises. No external dataset or executable code is required to establish the stated theorems.

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

References

[1] J. E. Hopcroft, R. Motwani, and J. D. Ullman, *Introduction to Automata Theory, Languages, and Computation*, 3rd ed., Pearson, 2006.

[2] R. P. Stanley, *Enumerative Combinatorics, Vol. 1*, 2nd ed., Cambridge University Press, 2012.

[3] G. Winskel, "Event structures," in *Petri Nets: Applications and Relationships to Other Models of Concurrency*, LNCS 255, Springer, 1987, pp. 325–392. DOI: 10.1007/3-540-17906-2_31.

[4] V. Pratt, "Higher dimensional automata revisited," *Mathematical Structures in Computer Science* 10(4) (2000), 525–548. DOI: 10.1017/S0960129500003169.